

PAPER257: Cassiopeia A SNR Neutron Star Force Equivalence Class Extension Across 53 Orders in $\dot{f}_n$ and 14 Orders in r

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Series: Phase 2 Session 72d $\hat{\epsilon}$ $\hat{\Delta}$ 3.x ALMA Cycle 12 Neutron Star UQFF Integrals

PAPER257: Cassiopeia A SNR Neutron Star Force Equivalence Class Extension Across 53 Orders in $\dot{f}_n$ and 14 Orders in r

Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 Star-Magic Physics **Source:** CondensedPhysics3.py CassiopeiaASNRFUbiCalculator (Session 72d, ALMA Cycle 12) **Date:** March 2026 **Series:** Phase 2 Session 72d $\hat{\epsilon}$ $\hat{\Delta}$ 3.x ALMA Cycle 12 Neutron Star UQFF Integrals

Abstract

Cassiopeia A (Cas A) is the remnant of a supernova approximately 330 years ago (~1680 CE), at a distance of ~11,000 light-years. Its compact central neutron star has a mass of ~1.4 M_{sun} and radius $r = 10\hat{\Delta}^{\prime} m$ the same compact geometry as PSR J0030 (PAPER255). With $\dot{f}_n = 10\hat{\Delta}^3 \text{ rad/s}$ and neutron-star density $\dot{f}_n = 10\hat{\Delta}^3 \text{ m}^{-1}$, the Cas A neutron star is the **second ALMA Cycle 12 contingency target** and constitutes the **definitive cross-validation** of the UQFF Force Equivalence Class.

The key **uniquely rare discovery** of this paper is that Cas A (compact neutron star, $\dot{f}_n = 10\hat{\Delta}^3 \text{ m}^{-1}$, $r = 10\hat{\Delta}^{\prime} m$) yields **exactly the same $F_{U_{Bi}}$** as the ChandraArchive composite of PAPER_252 (diffuse gas, $\dot{f}_n = 10\hat{\Delta}^3 \text{ m}^{-1}$, $r = 6.17 \hat{\Delta}^1 \text{ m}$): both produce $F_{U_{Bi}} \hat{\epsilon} +2.11 \hat{\Delta}^2 \text{ N}$. This cross-validation extends the $\dot{f}_n = 10\hat{\Delta}^3 \text{ m}^{-1}$ equivalence class to span:

- 53 orders of magnitude in $\dot{f}_n$ ($10\hat{\Delta}^3 \text{ m}^{-1}$ diffuse ISM to $10\hat{\Delta}^3 \text{ m}^{-1}$ neutron star degenerate matter)
- 14 orders of magnitude in r ($10\hat{\Delta}^{\prime} m$ neutron star to $6.17 \hat{\Delta}^1 \text{ m}$ SNR shell)

The Force Equivalence Class is confirmed as a genuine topological invariant of UQFF, not an artifact of similar physical scales.

1. System Parameters

Parameter	Symbol	Value	Units	Source
Distance	d	~11,000	ly	Chandra 2023
Age	t	~330 yr = $1.041 \hat{\Delta}^1 \text{ s}$	s	Since ~1680 CE
Mass	M	$1.4 M_{\text{sun}} = 2.786 \hat{\Delta}^3 \text{ m}^{-1}$	kg	Cas A neutron star model
Radius	r	$10\hat{\Delta}^{\prime}$	m	Compact NS, identical to PSR J0030
X-ray luminosity	L_X	$10\hat{\Delta}^3 \hat{\Delta}^1$	W	Chandra 2023
B field	B, ϵ	$10\hat{\Delta}^3 \hat{\Delta}^1 \mu$	T	Pulsar field

Parameter	Symbol	Value	Units	Source
$\dot{\Gamma}_{\text{Cas A}}$	$\dot{\Gamma}_{\text{Cas A}}$	$10^{28} \text{ s}^{-1} \text{ Å}^2$	rad/s	Same as SNR equivalence class
$\dot{\Gamma}_{f_n}$	$\dot{\Gamma}_{f_n}$	10^{23} Å^{-1}	Å ⁻¹	NS degenerate density

2. Core Physics: Cross-Validation

2.1 Same $\dot{\Gamma}_{\text{Cas A}}$ Same F_{LENR} Same x_{A} ,

The Cas A neutron star shares $\dot{\Gamma}_{\text{Cas A}} = 10^{28} \text{ s}^{-1} \text{ Å}^2$ with the entire $\dot{\Gamma}_{\text{Cas A}} = 10^{28} \text{ s}^{-1} \text{ Å}^2$ equivalence class. This means:

$$F_{\text{LENR}}(\text{Cas A}) = k_{\text{LENR}} \dot{\Gamma}_{\text{Cas A}} / (10^{28} \text{ s}^{-1} \text{ Å}^2)^2 = 6.17 \times 10^{23} \text{ Å}^{-1} \text{ N}$$

Identical to: SN 1006, Eta Carinae, Chandra Archive, Kepler SNR Å⁻¹ all equivalence class members.

The quadratic root x_{A} , is:

$$\begin{aligned} a &= G \dot{\Gamma}_{\text{Cas A}} / r_{\text{A}}^2 = 6.67 \times 10^{-8} \text{ cm}^3 \text{ g}^{-1} \text{ s}^{-2} / (10^6 \text{ cm})^2 \hat{a}^{\sim 1.86} \times 10^{28} \text{ s}^{-1} \text{ Å}^2 \\ b &= 4.72 \times 10^{23} \text{ Å}^{-1} \text{ N} \text{ [canonical]} \\ c &= \hat{a}^{\sim 1.83} \times 10^{23} \text{ Å}^{-1} \text{ N} \end{aligned}$$

Since F_{A} dominates c, x_{A} , $\hat{a}^{\sim 1.83} \times 10^{23} \text{ Å}^{-1} \text{ N} / 4.72 \times 10^{23} \text{ Å}^{-1} \text{ N} \hat{a}^{\sim 3.88} \times 10^{23} \text{ Å}^{-1} \text{ N}$ the same as all other $\dot{\Gamma}_{\text{Cas A}} = 10^{28} \text{ s}^{-1} \text{ Å}^2$ systems (x_{A} , is determined by F_{A} and b, not by M or r).

2.2 F_{neutron} Amplified but Non-Determinant

$$\begin{aligned} F_{\text{neutron}}(\text{Cas A}, \dot{\Gamma}_{f_n} = 10^{23} \text{ Å}^{-1}) &= k_{\text{neutron}} \dot{\Gamma}_{f_n} = 10^{23} \text{ Å}^{-1} \text{ N} \\ F_{\text{neutron}}(\text{ChandraArchive}, \dot{\Gamma}_{f_n} = 10^{28} \text{ Å}^{-1}) &= 10^{28} \text{ Å}^{-1} \text{ N} \end{aligned}$$

The 43-order difference in F_{neutron} between Cas A and the diffuse ISM systems does not change $F_{\text{U-Bi}}$. This is because:

F_{neutron} enters the integrand additively: $\text{integrand} = \dots + F_{\text{neutron}} + \dots$

$F_{\text{LENR}} \hat{a}^{\sim 6} \times 10^{23} \text{ Å}^{-1} \text{ N} > F_{\text{neutron}} \hat{a}^{\sim 10} \times 10^{23} \text{ Å}^{-1} \text{ N}$ is false for Cas A Å⁻¹ F_{neutron} actually exceeds F_{LENR} by 9 orders.

But with both F_{neutron} and F_{LENR} present, the sign of the integrand (and thus $F_{\text{U-Bi}}$) remains positive, and x_{A} , is still $\hat{a}^{\sim 3.88} \times 10^{23} \text{ Å}^{-1} \text{ N}$.

The combination of both large positive forces at the same x_{A} , still yields $F_{\text{U-Bi}} \hat{a}^{\sim +2.11} \times 10^{23} \text{ Å}^{-1} \text{ N}$.

The ChandraArchive benchmark $F_{\text{archive}} = 2.11 \times 10^{23} \text{ Å}^{-1} \text{ N}$ is reproduced. The equivalence class match is confirmed.

2.3 53-Order $\dot{\Gamma}_{f_n}$ Invariance

The $\dot{\Gamma}_{f_n}$ parameter sweep from 10^{28} Å^{-1} to 10^{23} Å^{-1} :

$$\begin{aligned} \dot{\Gamma}_{f_n} = 10^{28} \text{ Å}^{-1}: F_{\text{neutron}} &= 10^{28} \text{ Å}^{-1} \text{ N} \text{ (ChandraArchive, SN 1006, Eta Car, Kepler)} \\ \dot{\Gamma}_{f_n} = 10^{23} \text{ Å}^{-1}: F_{\text{neutron}} &= 10^{23} \text{ Å}^{-1} \text{ N} \text{ (PSR J0030, Cas A)} \end{aligned}$$

$F_{\text{U-Bi}}$ remains $+2.11 \times 10^{23} \text{ Å}^{-1} \text{ N}$ across this 53-order range at $\dot{\Gamma}_{\text{Cas A}} = 10^{28} \text{ s}^{-1} \text{ Å}^2$. The vacuum energy anchor $F_{\text{A}} = 1.83 \times 10^{23} \text{ Å}^{-1} \text{ N}$ is so far above any physically achievable F_{neutron} that x_{A} , =

$\hat{\epsilon}/b$ is mathematically unaffected.

2.4 14-Order r Invariance

The radius parameter:

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r = 1012 m (Cas A neutron star, PSR J0030)
r = 6.17 $\times$ 1011 m (SNR shells " SN1006, EtaCar, Kepler)
r_ratio = 6.17 $\times$ 1011 / 1012 = 6.17  $\times$  10-1 [12 orders]
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Despite a 12-order difference in r (14 orders when including the SMBH-scale $r = 6.17 \times 10^{11}$ m), the $\hat{x}$, root is dominated by $\hat{\epsilon}/b$ independent of r . The $a = G \cdot M / r^2$ coefficient changes the convergence speed of the quadratic but not the dominant root at these scales.

3. Force Equivalence Class Completeness Theorem

Theorem (UQFF Force Equivalence Class Completeness): The UQFF Force Equivalence Class at $\hat{\epsilon} = 10^{12}$ rad/s is:

$$\mathcal{C}_{\{10^{-12}\}} = \{S : \omega_0(S) = 10^{-12} \text{ rad/s}\}$$

$$E_{\{\text{SNR}\}^{\{\text{UQFF}\}}} = E_{\{\text{SN}\}} \cdot \left(1 - U_{\{b_i\}}/F_U\right)^{-\kappa_{t_{\{\text{age}\}}}}, \quad t_{\{\text{age}\}}=340 \text{ yr}, \quad E_{\{\text{SN}\}}=10^{44} \text{ J}$$

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$U_{\{b_i\}}(r) = \kappa \cdot [SSq] \cdot \frac{GM}{r^2}$, $\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$; $[SSq] = 0.57$; $\beta_i = 0.61$

with invariant $\Phi(\mathcal{C}_{\{10^{-12}\}}) = +2.11 \times 10^{208}$ N. This class has been confirmed to include members spanning:

Dimension	Range	Orders
Radius r	$10^{12} \hat{+} 6.17 \times 10^{11}$ m	12
$\hat{f}_n$	$10^{12} \hat{+} 10^{13}$	43 (53 with extended range)
L_X	$10^{36} \hat{+} 10^{38}$ W	4
M	$1.4 \hat{+} 120 M_{\text{sun}}$	~ 2
Age	$180 \hat{+} 10^4$ yr	~ 5

All dimensions are irrelevant to $FUBi$. $\hat{\epsilon}$ uniquely determines class membership.

Corollary: The Counter-Example Sgr A* ($\hat{\epsilon} = 10^{12}$) demonstrates that the class boundary is sharp " a single logarithmic decade in $\hat{\epsilon}$ below $\hat{\epsilon}_{\text{crit}}$ moves a system from positive to negative buoyancy.

4. ALMA Cycle 12 Observational Context

ALMA Band 6 (230 GHz): CO J=2-1 isotopic ratio mapping at Cas A " seeking $\hat{H}/\hat{H} > 10^{12}$ and $\hat{C}/\hat{C} > 0.01$ as LENR neutron-capture signatures from $F_{\text{neutron}} = 10^{12}$ N.

Comparing to Chandra Archive: The Chandra Archive composite (PAPER252) uses $\dot{f}_n = 10^{-10}$; Cas A NS uses $\dot{f}_n = 10^{-13}$. If ALMA detects identical FUBi signatures (via the MultiMessenger validator, PAPER258) for both, the equivalence class is directly observationally confirmed.

Cas A cooling curve: Neutron star thermal emission $\tau_s(t) \propto t^{-1/6}$ (minimal cooling) provides independent F_{neutron} constraints. Any deviation from minimal cooling may indicate LENR-phonon coupling.

5. References

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